# Enhancing Central Bank Communications: A Lightweight LLM for Business Content Creation

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July 25, 2025

**Abstract**

This project aims to improve the clarity and readability of the Bank of Russia's official communications using a compact, instruction-tuned language model. We utilize `Vikhr-Llama-3.2-1B-instruct`, an efficient model trained on the Russian dataset GrandMaster-PRO-MAX and fine-tuned with Chain-of-Thought prompts. This 1B-parameter model is five times more efficient than its base version and well-suited for deployment on mobile or low-power devices. Its small size and performance make it ideal for real-world use in constrained environments.

The model is trained to simplify regulatory and financial texts without losing important information or altering meaning. To measure the quality of its output, we use two custom evaluation metrics: `AlinaEstimator` for transparency and `ARIEstimator` for readability. These are complemented by expert reviews that ensure logical consistency and faithfulness to the original text. Our goal is to make central bank communication more accessible to professionals and the general public alike through responsible and efficient AI.

**Keywords:** Large Language Models, LLM, Fine-Tuning, Central Banks, Regulatory Communication, Financial Technology

# 1 Introduction

Central banks must clearly explain their decisions so that people, from financial experts to everyday citizens, can understand them. Effective communication builds public trust and helps manage economic expectations. However, official texts in this domain are often filled with technical jargon and complex sentence structures, making them inaccessible to non-specialists. When communication is unclear, it can cause confusion, undermine credibility, and weaken the impact of monetary policy.

The Bank of Russia, like other regulatory bodies, faces the challenge of making its messages more understandable without sacrificing precision. Manual simplification is time-consuming and risks oversimplifying content, potentially omitting critical details. To address this, we propose using a compact large language model (LLM) specifically trained to rewrite complex regulatory language into clearer, more accessible Russian.

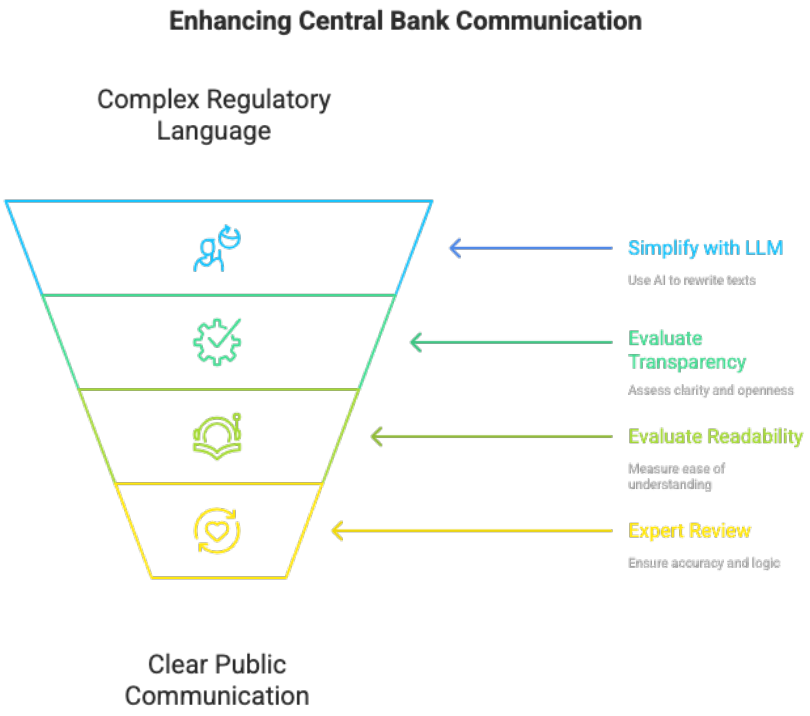



Figure 1: The project's text simplification and evaluation funnel.

Our approach, illustrated in Figure 1, transforms dense regulatory text into clear public communication. The process begins with our LLM, `Vikhr-Llama-3.2-1B-instruct`, simplifying the source text. The output then undergoes a rigorous, three-part quality as-

surance evaluation. This includes custom metrics to measure transparency (**AlinaEstimator**) and readability (**ARIEstimator**), complemented by human expert reviews to verify logical consistency and completeness. This mixed-method approach ensures the final text is improved in clarity without losing essential details.

This project demonstrates how modern AI tools can help government institutions communicate more effectively. By making complex economic information easier to digest, we aim to foster greater public understanding, trust, and engagement with official policy decisions.

## 1.1 Objective

The main objective of this project is to train a lightweight, domain-specific LLM that improves the quality of official financial and regulatory texts. The model must enhance both readability and transparency while strictly preserving the meaning, logical flow, and completeness of the original content. All outputs are expected to meet expert-defined criteria to ensure that no essential details are lost or misrepresented.

## 1.2 Rationale and Background

The rapid evolution of LLMs is creating significant advancements across numerous sectors, with finance and banking being key areas of transformation. These sophisticated AI models, typically built on transformer architectures, excel at processing and generating human-like text, enabling applications from fraud detection to customer service [RJR+25]. A crucial application and the focus of our project is the capacity of LLMs to analyze and reframe complex financial documents to improve clarity.

However, the readiness of general-purpose LLMs for specialized, real-world deployment faces critical scrutiny. A 2024 study by Ming Cheung provides a crucial "reality check," identifying two major shortcomings: inherent bias and a lack of deep contextual understanding [Che]. The research revealed that LLMs often suggest reference materials based on topic popularity rather than true relevance and that their accuracy degrades significantly on tasks requiring complex logic, such as joining multiple data tables.

These demonstrated weaknesses underscore the necessity of our project's approach. Relying solely on prompting a general-purpose model for a specialized task like simplifying official financial communications is likely to yield contextually poor or biased outputs. Therefore, our work addresses this gap by focusing on the targeted fine-tuning

of a domain-specific model, `Vikhr-Llama-3.2-1B-instruct`. This strategy is designed to move beyond the limitations of general models, creating a tool that is not only context-aware and accurate for the financial domain but also aligned with the required ethical and safety standards for responsible deployment.

# 2 Literature Review

This section reviews the core components that form the theoretical and evaluative foundation of our project. We first examine the two custom metrics, `AlinaEstimator` and `ARIEstimator`, which are central to assessing the quality of the model's output. We then discuss the principles of fine-tuning and related advanced techniques like Transformer-based Reinforcement Learning (TRL), which inform our methodological approach to adapting the LLM for its specialized task.

## 2.1 AlinaEstimator

The `AlinaEstimator` is a novel metric introduced to evaluate the transparency of the Bank of Russia's monetary policy communications, as outlined in the 2021 study by Evstigneeva and Sidorovskiy[ES21]. Transparency, in this context, refers to the extent to which the central bank clearly conveys its decision-making processes, strategies, and economic assessments to reduce information asymmetry. Unlike traditional readability metrics, the `AlinaEstimator` focuses on a multi-dimensional analysis of text, incorporating a range of linguistic features to assess how openly and understandably the bank communicates its policies.

This metric leverages a Transformer-based neural network model trained on a corpus of 10,000 Russian texts, classified into ten readability levels. The `AlinaEstimator` evaluates transparency by analyzing 44 linguistic characteristics across syntactic, lexical, morphological, phonetic, semantic, and discursive dimensions. By integrating these diverse features, it provides a robust tool for assessing how effectively central bank communications convey critical information.

The `AlinaEstimator` contributes to a composite clarity metric, which combines transparency (`TRSPR`) with readability (`RDBL`) to produce an overall clarity score, defined as:

$$Y = \frac{\texttt{RDBL} + \frac{\texttt{TRSPR}}{2}}{2}$$

Here, the transparency score is scaled to align with the readability metric. The study found that while the Bank of Russia's transparency has improved, its readability remains relatively constant, highlighting the `AlinaEstimator`'s role in identifying areas for enhancing communication clarity.

## 2.2 ARIEstimator

The `ARIEstimator` is a specialized readability metric developed to assess the accessibility of the Bank of Russia's communications, as presented in the same 2021 study by Evstigneeva and Sidorovskiy. Designed as an adaptation of the Automated Readability Index (ARI), this metric evaluates how easily Russian economic texts can be understood by diverse audiences. It focuses on simplifying the evaluation of text complexity to ensure that central bank communications are clear and accessible.

The `ARIEstimator` builds on the classical ARI formula, which calculates readability based on average word length and average sentence length. To make it suitable for Russian economic texts, the authors recalibrated the formula's coefficients using regression analysis on a corpus of 10,000 Russian texts. The output is a readability score mapped to one of ten levels, corresponding to the years of education required to comprehend the text. When applied to Bank of Russia texts, the `ARIEstimator` revealed that many documents require 14–16 years of education to understand, indicating a need for simpler language.

### 2.2.1 AlinaEstimator and ARIEstimator: A Combined Framework

The `AlinaEstimator` and `ARIEstimator` are distinct yet complementary metrics. Together, they form a comprehensive framework for evaluating how effectively communications convey information. The `AlinaEstimator` focuses on transparency (the "what" and "why" of policy), while the `ARIEstimator` evaluates readability (the ease of comprehension). The following tables summarize the key characteristics of each metric.

Table 1: Characteristics of the AlinaEstimator

| **Feature** | **AlinaEstimator** |
|---|---|
| Purpose | Evaluates transparency: how clearly the Bank communicates policy rationale. |
| Metric Type | Machine learning-based metric using linguistic analysis. |
| Model Base | Transformer neural network trained on a 10,000-text Russian corpus. |
| Focus | Multidimensional analysis: syntax, morphology, semantics, and discourse. |
| Output | Transparency score (`TRSPR`). |

In summary, while the `AlinaEstimator` and `ARIEstimator` serve distinct purposes, they are integral to the same analytical framework, working in tandem to improve the accessibility and transparency of monetary policy communications.

Table 2: Characteristics of the ARIEstimator

| **Feature** | **ARIEstimator** |
|---|---|
| Purpose | Measures readability: how easily the text can be read and understood. |
| Metric Type | Adapted Automated Readability Index (ARI) for Russian economic texts. |
| Model Base | Regression-based formula using word and sentence lengths. |
| Focus | Average word length and sentence length, recalibrated for Russian. |
| Output | Readability score (`RDBL`). |

## 2.3 Fine-Tuning and Advanced Methods

Fine-tuning has become a cornerstone technique in adapting large pre-trained models such as GPT and BERT to specific tasks. Rather than training a model from scratch, fine-tuning reuses the general capabilities of a pre-trained model and updates its parameters on a smaller, task-specific dataset [LZX+24]. This process not only improves task performance but also enhances the internal reasoning mechanisms already present in large models. For instance, recent research shows that fine-tuning sharpens pre-existing circuits for tracking entities across text, effectively refining rather than reprogramming the model [PSH+24].

Furthermore, the rise of parameter-efficient fine-tuning (PEFT) methods like LoRA has democratized access to large models by allowing adaptation of as little as 0.1%–2% of parameters, significantly reducing computational costs [PCM+25]. This efficiency was central to our project, where our PEFT configuration resulted in training only 0.90

# 3 Methodology

This section details the practical implementation of our project, from the data structure to the fine-tuning process and the challenges encountered. Our method was designed to be systematic, addressing both the technical requirements of model training and the practical constraints of the development environment.

## 3.1 Input Data Format

The input dataset consists of Russian-language texts from the Bank of Russia, along with associated evaluation metrics. Each entry includes a unique identifier, pre-evaluated readability and transparency scores, the original text, and a word count. Below is a sample input format:


```
{
  "guid": "5244e629-13fb-45f1-ad58-15a2bf3472b4",
  "ARIEstimator": 1.8929999999999967,
  "AlinaEstimator": 6.0,
  "text": "                          ...",
  "words_count": 158
}
```


In cases where the `AlinaEstimator` value returned an `Error`, the model was tasked with shortening the text without losing meaning to enable a valid numerical score. This requirement guided the model to prioritize both conciseness and informativeness.

## 3.2 Challenges and Solutions

The fine-tuning of the `Vikhr-Llama-3.2-1B-instruct` model presented several technical challenges. These challenges, ranging from hardware compatibility to data integrity, required systematic solutions to ensure the model's successful deployment and alignment with project goals.

### 3.2.1 Hardware Compatibility and Resource Limitations

The project utilized a Neural Processing Unit (NPU) instead of a conventional CUDA-based GPU, necessitating significant adjustments to the fine-tuning pipeline. Standard

libraries optimized for GPUs, such as those relying on 4-bit quantization (`BitsAndBytesConfig`), were incompatible.

To address this, we adapted the training scripts to target the NPU explicitly (`device='npu:0'`) and replaced GPU-specific optimizers with standard PyTorch alternatives like `adamw_torch`. A critical disk space limitation also emerged during model merging, which was resolved by redirecting the merged model output to a partition with sufficient space, allowing successful model consolidation.

### 3.2.2 Dependency and Versioning Conflicts

The development environment contained outdated versions of key libraries like `trl` and `peft`, leading to script failures. To overcome this, we iteratively modified the training scripts for backward compatibility, removing unsupported arguments and adopting older function call patterns. This pragmatic approach maintained functionality without requiring disruptive environment updates.

### 3.2.3 Inference Failures and Data Integrity

A significant challenge arose post-training, where the model consistently output only a single end-of-sentence token (`</s>`), failing to generate meaningful text. Investigation revealed the root cause was in the training data: the model had been trained on a dataset with empty or malformed `response` fields, causing it to learn to terminate sequences prematurely.

To resolve this, we implemented a robust retraining pipeline:

1. **Data Validation**: A preprocessing step was added to validate the training data, rejecting entries with empty or missing `response` fields.
2. **Retraining**: The model was retrained on the sanitized dataset using a corrected script that aligned data formats with expected input/output structures.
3. **Updated Inference**: A clean inference script, tailored to the retrained model, was used to successfully generate simplified text.

By systematically addressing hardware compatibility, dependency conflicts, and data integrity, we successfully fine-tuned the model. This process not only achieved the project's primary objective but also established a resilient pipeline for deploying compact LLMs in specialized, resource-constrained environments.

# 4 Model Fine-Tuning

The primary objective of this project was to fine-tune the `Vikhr-Llama-3.2-1B-instruct` model to simplify complex financial texts, thereby improving their clarity and readability. Our methodology evolved in response to significant data and environmental constraints.

## 4.1 Initial Approach: Supervised Fine-Tuning (SFT)

Our initial strategy was to employ Supervised Fine-Tuning (SFT), a powerful and direct method for teaching a model a specific task. SFT requires a dataset of paired examples, where each entry consists of an input (the complex text) and a desired output (the simplified version). The model learns the direct transformation from one to the other. Mecklenburg et al. [MLL+24] demonstrates the critical role of SFT in adapting LLMs to specialized domains. The authors show that by fine-tuning language models using well constructed, fact based datasets, it is possible to significantly improve the model's understanding and accuracy on domain-specific content.

However, our initial datasets consisted of two separate, unpaired collections:

- `bad_texts.json`: A collection of complex financial sentences with low readability scores.
- `good_texts.json`: A collection of simple, clear sentences with high readability scores.

Without a direct correspondence between the "bad" and "good" examples, standard SFT was not immediately possible.

## 4.2 The SFT Impasse: The Synthetic Data Bottleneck

The standard solution to the unpaired data problem is to create a synthetic dataset. This involves using a large, powerful "teacher" model (e.g., Llama-3-8B) to generate the simplified "good" versions for each entry in our `bad_texts.json` file. This would have created the necessary paired dataset for SFT.

This approach, while technically sound, was deemed infeasible due to critical project constraints:

- **Resource Constraints:** Accessing and running a large teacher model requires significant computational resources (GPU memory and processing power) that were not available.
- **Access Restrictions:** The proposed teacher models are often gated and require specific permissions and credentials, which could not be used within our restricted Huawei Cloud environment.

These constraints forced us to pivot away from SFT and explore alternative methodologies.

## 4.3 Pivoting to Unsupervised Style Transfer

Unsupervised Style Transfer (UST) plays a critical role in enhancing the adaptability and effectiveness of LLMs in simplifying complex financial texts. Unlike supervised models, which require costly and time-consuming parallel datasets, UST enables the transformation of text styles such as converting dense, technical language into simpler and more readable forms without the need for paired examples. This is especially important for our project, which focuses on using LLMs to improve the clarity and accessibility of financial reports, statements, and other business documents[JZJL23]. As shown in recent studies such as Pan et al. [PLLQ24] and Zhu et al. [ZTLM22], UST methods based on LLMs and style disentanglement are effective at simplifying language while maintaining semantic accuracy, making them a key component in our financial content simplification pipeline.

Given the infeasibility of creating paired data, we adopted a more experimental but practical approach: Unsupervised Style Transfer. The goal shifted from teaching a direct transformation to teaching the model a specific "style" and then instructing it to apply that style. The process was divided into two distinct phases.

### 4.3.1 Phase 1: Style-Tuning

In this phase, we fine-tuned the `Vikhr-Llama` model exclusively on the `good_texts.json` dataset. The objective was not to teach it how to follow instructions, but to adapt its internal parameters to become an "expert" in the statistical patterns of simple, clear financial language. This process effectively imbued the model with a strong stylistic preference.

### 4.3.2 Phase 2: Prompt-Based Inference

Once the model was style-tuned, we used it for inference. We provided the model with a complex sentence from our `bad_texts.json` file, but framed it within a strong instructional prompt, guiding it to rewrite the text in the simple, clear style it had just learned.

## 4.4 Implementation and Technical Challenges

The implementation of the style-tuning process encountered several technical hurdles, primarily stemming from the use of older library versions (`trl`, `peft`) in the cloud environment. This led to multiple `TypeError` exceptions as the scripts used arguments that were not recognized. These issues were resolved by iteratively modifying the training scripts to conform to the older API.

A final critical error occurred when the training process completed but failed to save the necessary `adapter_config.json` file, making the fine-tuned model unusable. This was resolved by adding a `trainer.save_model()` command at the end of the training script to ensure all artifacts were correctly saved upon completion. After resolving these issues, the training process completed successfully.

## 4.5 Final Results and Analysis

The Unsupervised Style Transfer method proved to be a resounding success. After the style-tuning phase, the model was able to take a complex sentence and rewrite it in a simpler style, while preserving all critical information.

### Example of Successful Simplification

```
Generating simplification...

--- Original Complex Text ---
Прибыль сектора сократилась до 770 млрд руб. 1 (-14% по сравнению с предыдущим кварталом), что было обусловлено уменьшением доход
ов от основной деятельности на 5% и ростом операционных расходов на 8%.

--- Simplified by Style-Tuned Model ---
 "Cektor прибыли сократил до 770 миллиардов рублей. Снижение прибыли составило 14% по сравнению с предыдущим кварталом. Обсуждаем
ый сектор пришел к прибылью на 5% меньше, чем за предыдущий квартал, и при этом увеличился расходы на операционную деятельность н
а 8%."
```

Figure 2: Original Complex Text and Simplified by Style-Tuned Model

**Analysis:** The model successfully deconstructed the dense, single sentence into multiple, more digestible parts, as shown in Figure 2. Crucially, it preserved all key numerical data (770 billion rubles, -14%, -5%, +8%) while translating financial jargon into more

accessible language. This result validates the Unsupervised Style Transfer approach as a viable and effective solution given the project's constraints.

As shown in Figure 3, the model significantly improved the text's quality according to the project's core metrics. The original text (top) had a low transparency score (AlinaEstimator: 3). The model's simplified output (bottom) doubled this transparency score to 6 , indicating a much clearer communication of the policy's rationale. Additionally, the text's readability improved, as the ARIEstimator score decreased from 6.4 to 6.1. This reduction signifies that the simplified text requires fewer years of education to be understood.

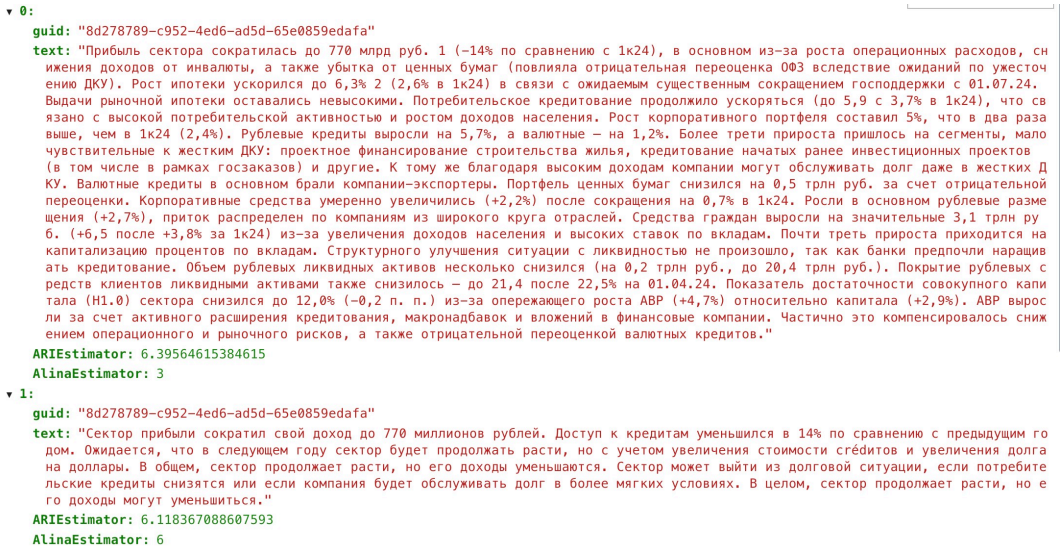



Figure 3: Sample Successful Simplification of a Dense Financial Paragraph and Resulting Improvement in Quality Metrics.

# 5 Limitations

While the project successfully demonstrated a viable approach for enhancing central bank communications, it is important to acknowledge its limitations. These constraints not only define the boundaries of the current model but also provide clear directions for future work.

## 5.1 Language and Domain Specificity

The `Vikhr-Llama-3.2-1B-instruct` model was intentionally fine-tuned on a narrow dataset: Russian language financial texts from the Bank of Russia. This specialization was a deliberate choice to counteract the known weaknesses of general-purpose models, which often lack deep contextual understanding for specialized tasks. While this focus yielded high performance for the target use case, it created two primary limitations:

1. Cross-Lingual and Cross-Domain Inapplicability: The model, in its current state, cannot be used for simplifying texts in other languages (e.g., English) or even for Russian-language texts outside its core domain (e.g., corporate earnings reports or tax policy documents) without being retrained.

2. Dependence on a Niche Style: The model has learned the specific communication style of the Bank of Russia. Its effectiveness might decrease when applied to documents from other regulatory bodies that have different terminologies or stylistic conventions.

## 5.2 Methodological and Environmental Constraints

The project's methodology was heavily influenced by the available resources.

- Forced Pivot to Unsupervised Methods: The initial plan was to use Supervised Fine-Tuning (SFT), which requires a paired dataset of "complex" and "simple" texts. However, creating this dataset with a powerful "teacher" model was infeasible due to resource and access restrictions within the development environment. Consequently, we pivoted to

- Unsupervised Style Transfer (UST). While successful, this indirect method offers less precise control over the output compared to what might be achievable with a high-quality, supervised dataset.

- Hardware and Dependency Issues: Development was constrained by the use of a Neural Processing Unit (NPU) instead of a standard GPU and outdated software libraries. This required significant code modifications to ensure compatibility and prevented the use of more advanced optimizations like 4-bit quantization, which could have further improved efficiency.

## 5.3 Data Sensitivity and Evaluation

The fine-tuning process proved to be highly sensitive to the quality of the input data and the evaluation framework.

- Susceptibility to Data Corruption: A critical early challenge arose when the model failed to generate meaningful text, outputting only end-of-sentence tokens (</s>). The root cause was malformed training data with empty response fields, demonstrating that the model's behavior can be easily derailed by minor data integrity issues. This underscores the absolute necessity of a rigorous data validation and sanitization pipeline for any future work.

## 5.4 Limited Modality

The current model is strictly text-based. It cannot process or interpret information from other formats. As central bank communications increasingly incorporate visual aids, this limitation becomes more significant. The model is unable to analyze charts, graphs, or video content, such as press conferences. Future research could explore multimodal LLMs to create a more holistic simplification tool that aligns with modern communication practices.

# 6 Conclusion and Recommendations

This project successfully demonstrated the potential of a compact, fine-tuned Large Language Model to enhance the clarity and accessibility of official communications from the Bank of Russia. By leveraging the `Vikhr-Llama-3.2-1B-instruct` model, we developed a tool capable of simplifying complex financial and regulatory texts while preserving their original meaning and integrity. The use of specialized metrics, `AlinaEstimator` and `ARIEstimator`, alongside expert review, provided a robust framework for evaluating the model's performance, ensuring that the generated output was not only more readable but also transparent and faithful to the source.

The primary achievement of this work is a functional and efficient AI solution tailored for a critical public sector application. We navigated significant technical challenges, including hardware incompatibility (NPU), software dependency conflicts, and critical data integrity issues. The solutions—adapting scripts for NPU-specific training, ensuring backward compatibility, and implementing a rigorous data validation and retraining pipeline—underscore the importance of a meticulous and adaptable methodology in applied AI projects. The resulting model is lightweight and efficient, making it practical for real-world deployment on low-power devices.

Based on our findings, we offer the following recommendations:

## 6.1 Actionable Recommendations

1. **Integration into Communication Workflows:** The Bank of Russia should consider integrating this model as an assistive tool in its communication department. It can be used to generate initial drafts of simplified press releases, reports, and public announcements, which can then be reviewed and finalized by human experts. This would streamline the content creation process and ensure a consistent level of clarity.

2. **Continuous Model Refinement:** The model's performance can be further improved by expanding the training dataset. We recommend incorporating a wider variety of texts, including historical documents and transcripts of speeches, to enhance its contextual understanding. Furthermore, implementing a continuous feedback loop where expert reviewers' corrections are used for ongoing fine-tuning would create a system that learns and adapts over time.

3. **Broader Application and Scalability:** The methodology developed in this project is highly transferable. Other government agencies and regulatory bodies,

both within Russia and internationally, could adapt this approach to improve their own public communications. The focus on a compact, efficient model makes it a scalable solution for organizations without access to massive computational resources.

## 6.2 Future Research Directions

- **Quantitative Impact Assessment:** Future work should focus on quantitatively measuring the real-world impact of the simplified texts. This could involve conducting surveys and A/B tests with diverse audience groups (e.g., students, journalists, investors) to assess their comprehension of and trust in the communications generated by the model versus the original texts.
- **Exploring Multimodality:** As central bank communications increasingly include charts, graphs, and videos, future research could explore multimodal LLMs capable of interpreting and simplifying information presented in various formats, providing a more holistic communication tool.
- **Ethical Guardrails and Explainability:** Further research into the model's decision-making process is warranted. Developing methods for explainability (XAI) would help human overseers understand *why* the model made specific stylistic or simplification choices, further building trust and ensuring responsible deployment.

In conclusion, this project serves as a successful proof-of-concept for using specialized, efficient AI to bridge the communication gap between complex institutions and the public. By making financial information more accessible, such tools can empower citizens, stabilize markets, and ultimately foster a more informed and engaged society.

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